

MSc Project Report
School of Engineering and Computer Science
University of Hertfordshire

Design of an electric 3-wheeler (adult) tricycle for countryside

Report by
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ABSTRACT

This project is a design development study of an electric tricycle for adults with minimal effort required by the driver. The study comprises work done in the spectrum of design conceptualization to CFD analysis of the design selected based on different criteria with respect to its driving environment. The scope of the project includes researching existing tricycle designs and materials, sourcing parts from third-party suppliers, and creating concept sketches. The selected design is modelled using CATIA V5 and analysed using computational fluid dynamics (CFD). Mesh Independency study of the model to attain the appropriate mesh size suitable for analysis is identified. Material optimization of the frame is performed by testing three varied materials under various loading conditions to determine the optimal material for the frame. The stress concentration points of the optimized frame are then identified, and the design is modified to better distribute stress and minimize deformation. The resulting design has the potential to serve as the foundation to produce adult electric tricycles operating under a load of around 4500kg.

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GLOSSARY

CFD: Computational Fluid Dynamics

FEA: Finite Element Analysis

TRIKE: Tricycle

C1: Concept 1

C2: Concept 2

CFRP: Carbon Fibre Reinforced Polymer

GAC: General Analysis Connection

A6061: Aluminium 6061 Alloy

ICM: Initial CAD Model

OCM1: Optimised CAD Model 1

OCM2: Optimised CAD Model 2

1. INTRODUCTION

Current means of transportation are either human-powered, powered by internal combustion engines, or powered by electric motors. The necessity to travel from one location to another has spawned its own business. While bicycles are among the healthiest modes of transportation, they lack the stability and comfort of a tricycle. They were initially designed to allow those with mobility issues to travel by using their own strength. The design has progressed tremendously over time. Tricycles have shown to be more practical than bigger vehicles that may be difficult to manage in tight settings, from child tricycles to freight tricycles used in industries to move heavy items in limited spaces.

1.1 ELECTRIC TRICYCLES



Figure 1: Trike electric bike from PEDEGO electric bikes

Tricycles have been designed so that the electric motor attached to the pedal or wheel hub generates additional power to propel the vehicle through which the energy needed by the driver is less than half and can commute for longer distances. Tricycles provides the best alternative for adults rather than a bicycle with advantages such as:

- Electric tricycles have three wheels instead of two which make them more stable than electric bicycles. This is especially beneficial for persons who have problems balancing on a standard bike or who prefer the increased stability of a tricycle
- With a lower centre of gravity and a larger wheelbase than electric bicycles, making them easier to drive

- Electric tricycles often have an electric motor to assist riders in pedalling, making cycling simpler and more pleasurable
- Riding a tricycle is environment friendly and is an effective method to stay healthy
- Electric tricycles can often carry greater weight and have more storage capacity than electric bicycles, making them an excellent choice for persons who need to move groceries, parcels, or other goods

Nevertheless, the disadvantages of electric tricycles are:

- The cost of an electric tricycle is significantly more expensive than bicycles
- They tend to be heavier which makes them difficult to transport
- Due to the extra weight and longer wheelbase, the manoeuvrability of an electric trike is constricted
- Heavier loads on the trike (tricycle) means less range of the electrical system
- Electric tricycles may be less common than standard bicycles, especially in specific places

1.2 AIM

This project aims to study and design an electric tricycle that can take three people with minimal effort from the driver on country roads. Geographical research, identifying necessary electrical components through literature research, design and optimising the model in CAD and Solidworks with varying effective parameters such as forces acting on the model, material optimisation is studied.

1.3 OBJECTIVES

The objectives of this project include,

- To perform comprehensive research on design of the tricycle, electrical components used for the final model with respect to range and payload
- Propose different concepts in relation to previous development attempts which are recognized through literature review
- Construct product design specifications which enlist all aspects of the tricycle needed to design and optimise
- Propose electrical components for the specific topology with respect to payload and range
- Create a detailed 3D model in CATIA V5 which mimics the model in real world
- Import the 3D model to the Finite Element Analysis workspace in CATIA to study the effects of changes on the design by varying aspects of force
- To validate the proposed design simulation (FEA, calculations, etc) using plots and other visual aids.

2. METHODOLOGY

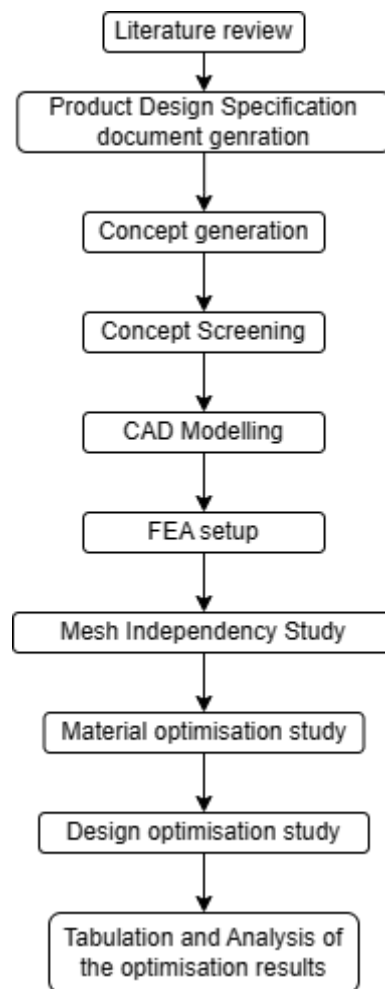


Figure 2: Work process flowchart

2.1. LITERATURE REVIEW

Arinze David et.al in their study of using electric tricycles for commercial transport in the region of Nigeria explains that when compared to typical petrol-powered tricycles, the usage of electric tricycles is a more cost-effective and ecologically beneficial solution. They reduce carbon emissions by thirty-two tonnes. Electric tricycles are also designed to be safer, with speeds limited to 30 km/hr and improved braking capabilities. This project intends to offer a reliable alternative source of electricity for cars in Nigeria, reducing dependency on gasoline and diesel and assuring the commercial transportation sector's viability despite possible challenges in the petroleum industry. ^[1]

Mutqin Abdul Muiz et.al conducted a static simulation study of tilting electric tricycle frame using FEA. Stress analysis of the frame was done to determine numerous factors such as Von-Mises stress, strain, displacement, and safety factor. Furthermore, they discovered that the minimal safety factor from the design simulation was 1.83 when mild steel was used as the

material with the main consideration being the distribution of static loads on the frame. Material Selection process of the study brought up three varied materials commonly used in bicycle frames namely, Aluminium 6061, Mild steel, and Carbon Fibre Reinforced Polymer (CFRP) whose properties are listed in the table below.

	Al 6061	Steel mild	CFRP
Elastic Modulus	68.9 GPa	220 GPa	133 GPa
Poisson's Ratio	0.33	0.28	0.39
Shear Modulus	26 GPa	82 GPa	53 GPa
Mass Density	2.7 g/cm ³	7.85 g/cm ³	1.75 g/cm ³
Tensile Strength	310 MPa	367 MPa	577 MPa
Yield Strength	276 MPa	306 MPa	550 MPa

Table 1: Material Properties (Mutqin Abdul Muiz et.al, 2020)

After the simulation study, CFRP displayed the best level of safety among the three materials evaluated (CFRP, aluminium 6061, and mild steel), with a minimum safety factor value of 2.64. The safety factor value for aluminium 6061 was 2.41, whereas the value for mild steel was 1.83. The lightest frame mass was 3.14 kg for CFRP, followed by 4.19 kg for aluminium 6061 and 6.92 kg for steel mild [2].

P.K. Devan et.al in their study of solar powered tricycle for physically challenged explained that the use of solar panels to power a vehicle has diverse benefits, including the fact that it is an ecologically benign means of transportation because it emits no greenhouse gas emissions. Hybrid tricycle is appropriate for those with physical limitations and may be used in both rural and urban locations. With a top speed of 20-25 kmph and is 30% more efficient than comparable electric and motorised tricycles. The cost of the unit might be decreased by 40-50% using mass manufacturing techniques. The use of hybrid tricycles can improve mobility for physically challenged people while also being a cost-effective mode of transportation [3].

Parameter	Solar powered tricycle	Electric tricycle	Hand powered tricycle	Motorized tricycle
Max. speed limit (km/hr)	20	20-25	15-20	25-30
Requirement of human Pedalling	No	No	Yes	No
Cost of the unit	25000	35000	6000	15000
Cost in rupees for operating the vehicle for 30km	Nil	Nil	Nil	75
Fuel used per 50km	Nil	Nil	Nil	1.25L
Type of energy used	Solar	Direct current	Muscle power	Petrol
Driving noise	Noiseless	Noiseless	Noiseless	70-80
Requirement of license	No	No	No	Yes
Age limit to operate the vehicle	No	No	No	Yes
Engine size	Not Applicable	Not Applicable	Not Applicable	100-110 cc

Table 2: Comparison of parameters for Solar powered, Electric, Hand powered and Motorized tricycle (P.K. Devan et.al, 2021)

Miroslav Blatnický et.al in their paper which is focused on improvement of driving properties of a tricycle structure with a steering system demonstrated a graph (Figure 3) was constructed using a 3D CAD model of the vehicle to demonstrate the link between the vertical position of the front wheel and steering wheel rotation. The graph shows that while the car is travelling straight, the front wheel is at zero height (the origin of the coordinate system is set this way). The height of the front component of the vehicle varies when the wheel spins and tilts, causing the position of the vehicle's centre of gravity to drop ^[4].

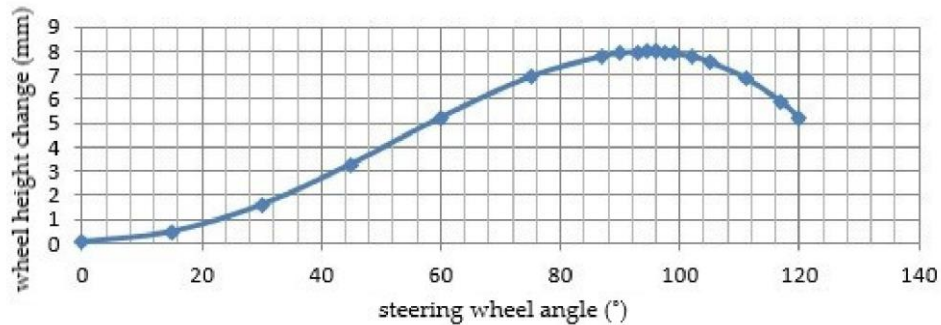


Figure 3: Front wheel height and steering wheel angle relation graph
(Miroslav Blatnický et.al, 2022)

Sukmaji Indro Cahyono et.al while performing finite element analysis of electric bicycle frame explained that the electric trike was created by reverse engineering pedicabs and public transit cycles. Several metal monocoque chassis with fibreglass composite inserts were evaluated, but they were not sturdy enough to withstand vibrations at the metal-composite junctions. The square tube was chosen because it was easier to manufacture and had a similar or greater size to the circular/oval tubes. The trike frame was heavier than a bike frame, and while the total frame weight rose only little when the diameter of the circular/oval tubes grew, it increased dramatically as the size of the square tube increased. The oval tube was the most secure.

The oval tube exhibited the highest safety factor and outperformed the circular tube under vertical forces. A variable three oval tube with diagonal dimensions of 56.6x40 mm and a thickness of 1.65 mm was the winning design. When compared to the circular tube, which was stronger against horizontal forces than the oval tube, this structure was robust enough to carry a maximum load of 700 kg and had a rapidly growing trend line for the factor of safety when the diagonal dimension was raised ^[5].

J. P. Srivastava et.al designed and constructed a human-electric hybrid power tricycle in response to the need for more sustainable transportation options. They believe that the current crisis in fuel energy consumption and conservation would lead to a reduction of the world's fuel reserves to 20% of their current level by 2040, so they wanted to create an alternative mode of

transportation. They created a three-wheeled vehicle with a Brushless DC 1.4 HP electric motor that is powered by four sealed 12-volt lead acid batteries and a chain drive. The materials used to construct the tricycle were chosen with the goal of minimising weight, strength, and cost ^[6].

2.2. PRODUCT DESIGN SPECIFICATION

Based on the literature survey, specifications for product design specification are recognized well within the working conditions in real world and are stated below:

Performance:

Motor: Pendix eMachine: **Gearless mid-mounted motor** fitted in contact with the pedal wheel. Max Performance of **250W** with max torque at **65Nm** capable of top speeds of up to **25km/h**. This motor is water and dust proof

Battery: Pendix ePower: The battery has a nominal voltage of **48V** and a capacity of **497Wh (10.4Ah)**. Weighing at **2.9Kg**, has an optimum range of **62-120km**. The battery can be charged within **3h 10min** with a charger that supports **160W**.

Gear mechanism: The pedal crank is connected to a **seven speed freewheel** at the rear set up on the rear axle along with a **7-speed tourney derailleur**. The gears on the bike can be changed using the **shifter** fitted on the handlebar

Size:

The size of the motor is 292 x 206 x 50mm connected to the cylindrical battery with dimensions (d x h) 80 x 342mm. The dimension of the tricycle is (l x w x h) 1800 x 1200 x 1500mm. The frame is connected to one tyre at the front of diameter 609.6mm (24") and two tyres at the rear with diameter of 508mm (10").

Frame:

The frame is constructed with **Aluminium 6061 alloy**. This can support loads up to **450kg** while weighing under **30kg**. It is designed to accommodate three people including the driver.

Safety: Fitted with reflectors for visibility, lights, bell or horn and mirrors.

Security: Locks for securing the trike to a fixed object.

Finish: Matte black finish on the frame with gloss black on the wheel rims.

Environment: Battery can be operated within (-10°C to 50°C)

Cost: Manufacturing cost plus £100

Quantity: Batches of 1000 – three thousand for domestic and international markets

Maintenance: Minimal (Lubrication for moving parts)

Weight: 35kg maximum

Product life: 15 years maximum

Shelf life: 10 years maximum

2.3. CONCEPT GENERATION

With the researched data and inspiration from the current market of tricycles, the concept drawings were made.

The prerequisite of having seating for three people (including driver), the designs show how the final design of the tricycle might look like meeting the requirement. Basic essential components like seats, bicycle frame, suspension, tyres to support the weight are mentioned in the drawings. The motor and the battery system (i.e., Pendix eDrive) chosen for the tricycle is considered while generating concepts so that it is included in the overall design with other components.

2.3.1. CONCEPT 1 (C1)

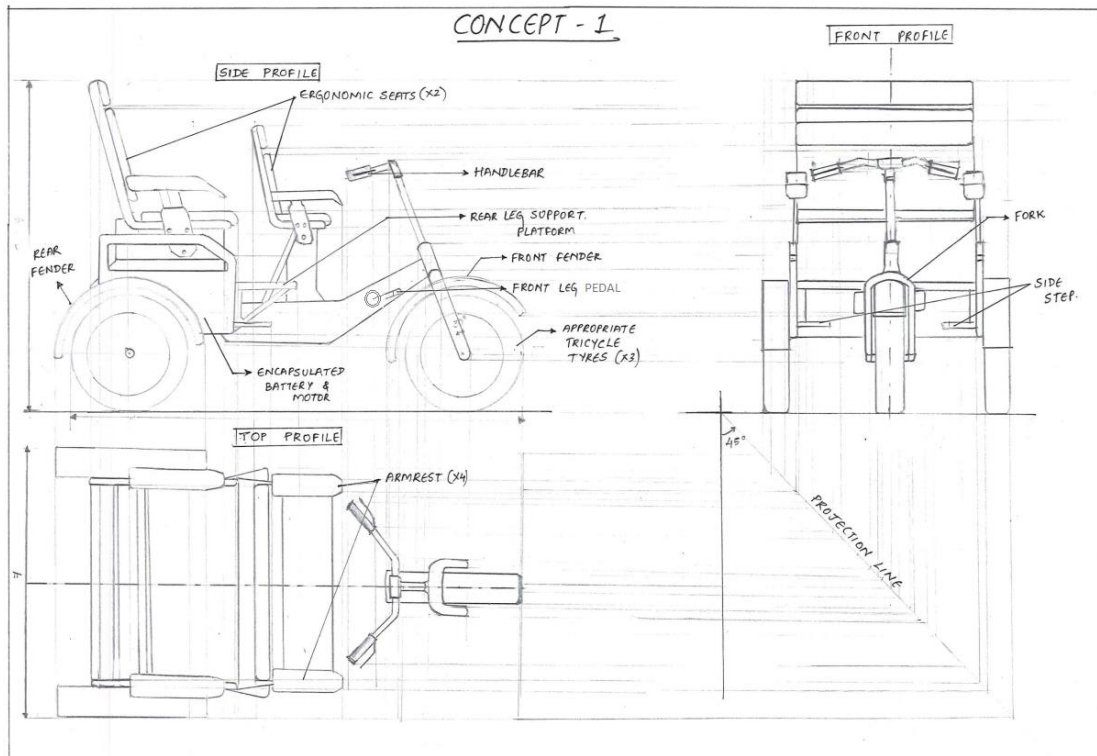


Figure 4: Concept 1 (C1) pencil sketch

The C1 sketch represents the one of the ideas of an electric tricycle. Space confinement of all the elements onto a frame without compromising the basic functionality of the vehicle was the main challenge. Representation of certain features in the sketch which are annotated contribute towards the process of resolving the challenge. Special comfort features like ergonomic seats to support the spine and leg rests to support people having trouble getting on the trike. The handlebar is positioned at an optimal height and position to reduce stress on the driver's body.

The frame of the tricycle encapsulates the working of motor and the battery which constitutes the electrical system of the tricycle.

2.3.2. CONCEPT 2 (C2)

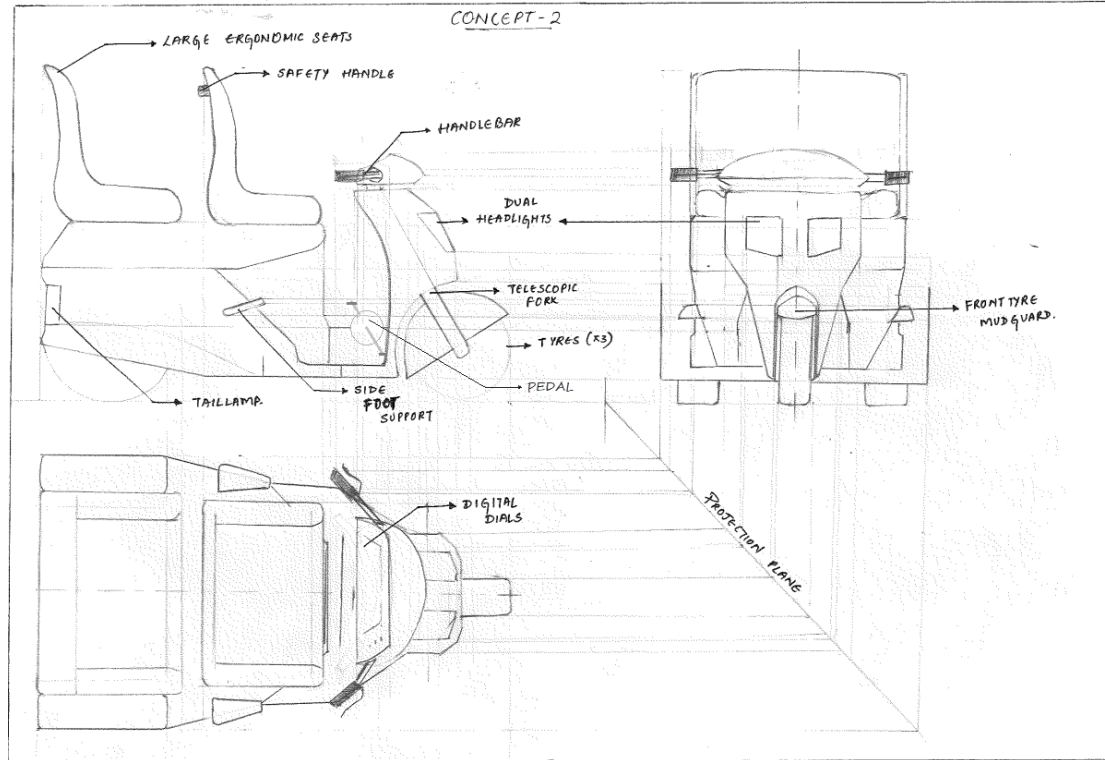


Figure 5: C2 pencil sketch

C2 concept sketch was inspired by old scooters but with the components of a pedal-electric system as same as C1. The body panels on C2 are much more refined than the naked design of the C1. With features like built-in headlights and taillamps, dials to show relevant data to the driver, etc, this is a design which would be an ideal electric tricycle. However, the screening process of the concepts introduced a new perspective of these vehicles.

2.4. CONCEPT SCREENING

Concept screening is a process of evaluating and comparing concepts in order to identify the most promising option. The method of concept screening used in this project is the Ranking method, where a score is assigned to each criterion such as:

- *Manufacturing feasibility:* The product's ability to be manufactured utilising current or future manufacturing techniques, technology, and resources within the specified cost and time restrictions
- *Technical feasibility:* Ability to be successfully implement the proposed solution using available resources and technology

- *Maintenance*: Doing certain activities to preserve the life of the product such as inspection, cleaning, repairing and replacing parts
- *Weight / Ease of handling*: The handling of the tricycle under various environments with less difficulty
- *Consumption of resources*: Amount of resources consumed to manufacture the product
- *Weight capacity*: Weight supported by the vehicle based on the strength of the materials used
- *Safety*: Safety features provided in the product
- *Emissions*: Gases or substances produced by the vehicle when it is running
- *Ease of Use*: Easiness in operating the vehicle
- *Style*: Physical design of the product

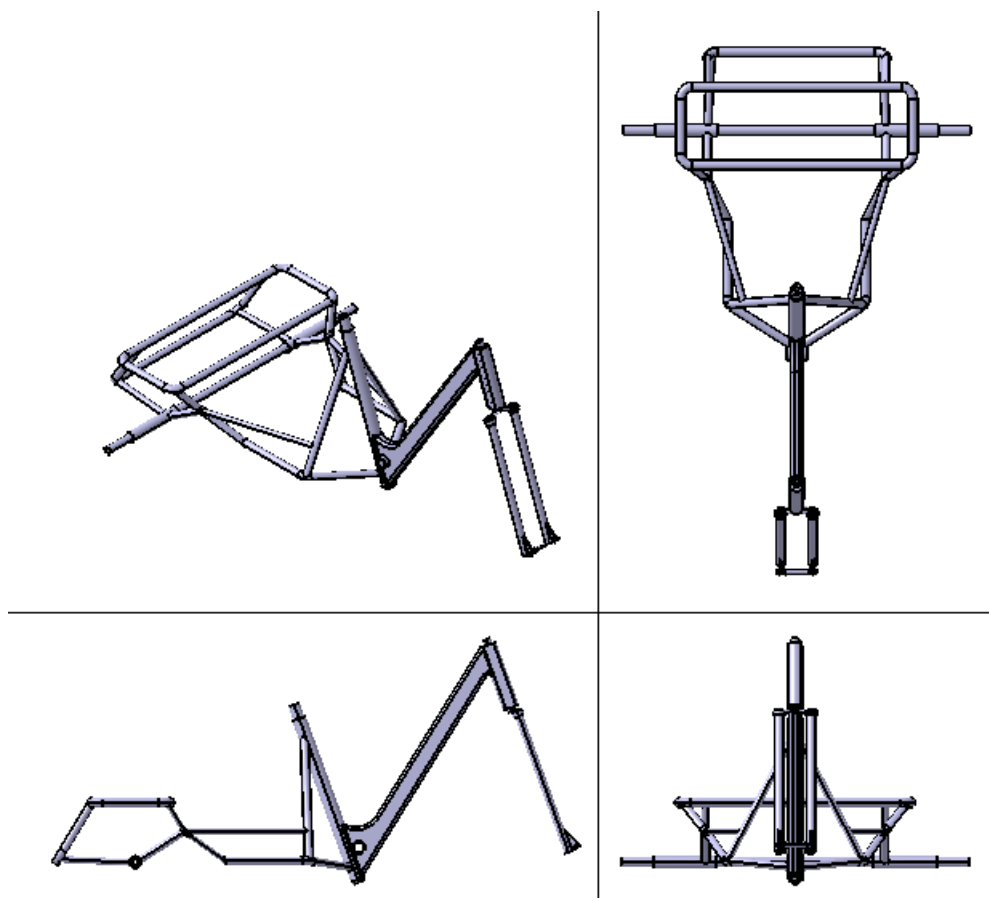
Comparison criteria	CONCEPT 1 (C1)		CONCEPT 2 (C2)	
Manufacturing feasibility	4		3	
Technical feasibility	3		3	
Maintenance	5		4	
Weight / Ease of handling	4		2	
Consumption of resources	3		3	
Transport capacity for goods	3		3	
Safety	3		3	
Pollutant emissions	5		5	
Ease of Use	4		4	
Style	3		4	
Total points	37		34	

Table 3: Design screening points table

Table 3 depicts the ranking system used for the screening process of the two concepts. The ranking or points in the table are based on the sketches developed and practicality. The C1 scores higher than C2 with better maintenance and ease of handling scores. This is because of the design of the C1 which does not have much body panels unlike C2. Maintenance of electric tricycle includes cleaning, inspection of electrical components, lubricating moving parts, etc. However, as the number of parts present are simple and attached onto the frame which are accessible. This makes the maintenance a bit easier on C1 than C2 due to the presence of body panels. The emission of the electric trike is less because of the vehicle is powered by electricity which produces zero emissions. Safety factors such as lights, reflectors on the frame and pedal, horn or bell, brakes, chain guards are some of the basic safety features that are attached with the design. Other criteria mentioned in the table are self-evident and support the results of the screening.

2.5. CAD MODELLING

The development of CAD model of the frame of the C1 is adopted from the concept sketch. Certain parameters of the frame such as the thickness of the frame, height, width and length of the tricycle had to be considered. Meeting the requirement of designing an electric tricycle for three people, frame structure with fixtures for the seats, wheels, motor, battery was made using CATIA V5. Frame, fork and axles are the three parts in the assembly used for the FEA analysis because the axles holding the tyres generates a descriptive figure during Finite Element Analysis of the applied forces on the frame. This can be used for the skeletal development of the trike.



*Figure 6: All view representation of the tricycle assembly
(Clockwise from top left: isometric, top, front, side view)*

2.6. FINITE ELEMENT ANALYSIS SETUP

Finite Element Analysis consists of procedures to obtain promising results that can be dependable. The 3D model of the assembly is imported into Generative Structural analysis workbench in CATIA V5. Mesh Independency study, Optimisation procedures are to be done using this workbench which are discussed in detail below.

2.6.1. MESH INDEPENDANCY STUDY SETUP

The environment for the mesh independency study is based on the software capability to mesh the CAD model and solve the load scenarios applied. These load scenarios are different from the ones which are applied during design optimisation techniques. The aim of the mesh independency study is to identify the most suitable mesh size for producing accurate results. The time taken to run the simulation is noted which helps the project meet its time constraint.

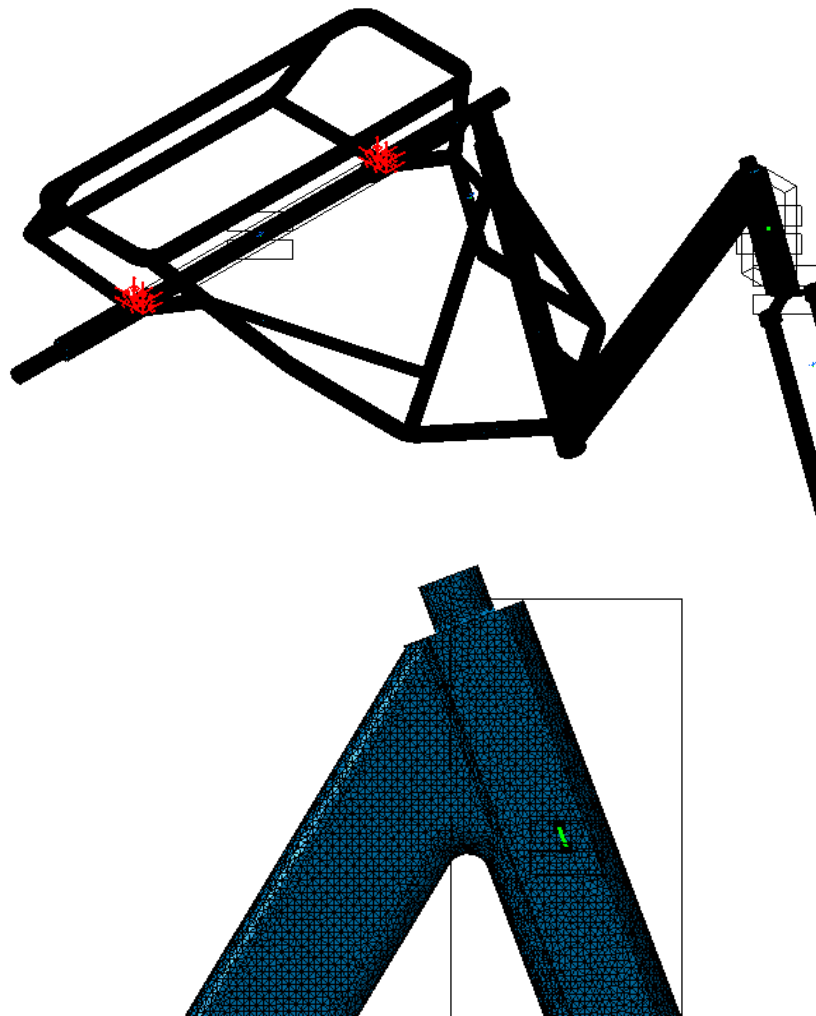


Figure 7: (Top) Mesh size depiction (Bottom) Meshed body

Different combinations of Mesh size and Sag size are used to the software to run the Static Analysis as shown in the table below. Conducting this analysis, we can identify the minimum required mesh size for accurate results.

Mesh Size (mm)	10	8	6	5	5	4	4	3	3
Sag (mm)	8	6	5	4.5	4	3.6	3	2.7	2

Table 4: Mesh size (mm) and Sag size (mm) combinations

2.6.2. CONFIGURING MATERIAL OPTIMISATION VARIABLES

As per Mutqin Abdul Muiz et.al ^[2], three materials commonly used in the cycle industry are Aluminium 6061 alloy, mild steel, Carbon-Fibre Reinforced Polymer. Implementing these materials for the CAD model we can decide the ideal material which can withstand the load with less deformation, inexpensive and dependable. The table below shows the data that was used to create a material library in CATIA V5.

Material	Young's Modulus (N_m2)	Poisson's Ratio	Density (kg_m3)	Yield Strength (N_m2)
Aluminium 6061 alloy	6.89E+10	0.33	2700	2.76E+08
Mild Steel	2.20E+11	0.28	7850	3.06E+08
CFRP	1.33E+11	0.39	1750	5.50E+08

Table 5: Material Data

These materials were particularly considered because of its extensive use in the production of bicycles. When applied with these properties per simulation on the CAD model, the analysis is conducted. The result of the analysis is explained in detail under [Results and Discussion](#) section below.

2.6.3. DEFINING DESIGN OPTIMISATION PARAMETERS

Design optimisation of the CAD model with the ideal material is conducted through Generative structural analysis tool in CATIA. The main objective of this optimisation is to reduce the amount of displacement of the frame and reduce the stress concentration points by smoothening it to provide a path on the frame for the stress to dissipate more evenly. The environment setup for the design optimisation consists of the following procedures:

- Connections: Defining analysis connections on the assembly to demonstrate the contact points on the assembly. GAC (General Analysis Connection) is applied to first between the fork and the frame. There are two contact points to be defined here. The third GAC is applied between the rear axle and the frame. (Figure 8)

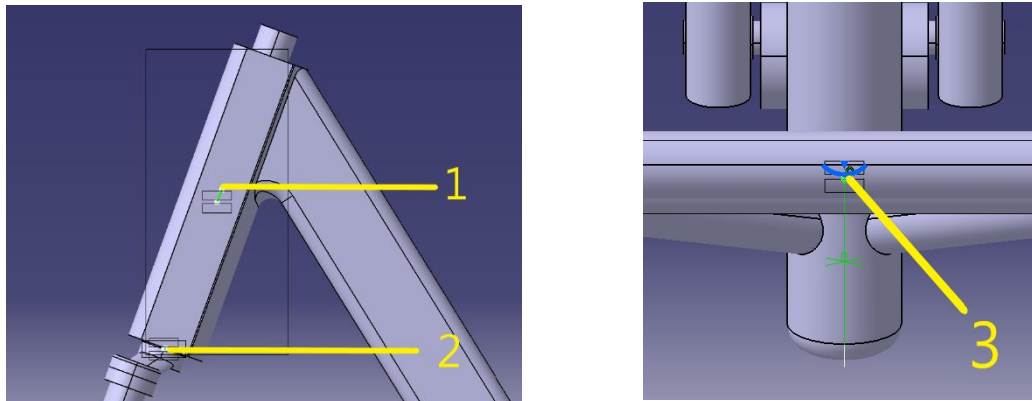


Figure 8: General Analysis connections 1, 2 & 3

- **Mesh Size:** The mesh size of all the elements in the assembly is changed to the ideal size which was considered during the mesh Independency analysis. Soon after defining the GAC's to the body, mesh visualization is applied for the contact meshes with respect to the connections to generate. This reduces the errors when applying restraints to restrict the model's movement in any direction. (Figure 9)



Figure 9: Meshed body

- **Restraints:** Restraints are applied on certain features of the assembly because we are conducting static analysis of the model. Therefore, some elements in the assembly should be fixed, in this case, the front and the rear axle. This is done to simulate the stationary position of the tyres while at a stop. (Figure 10).

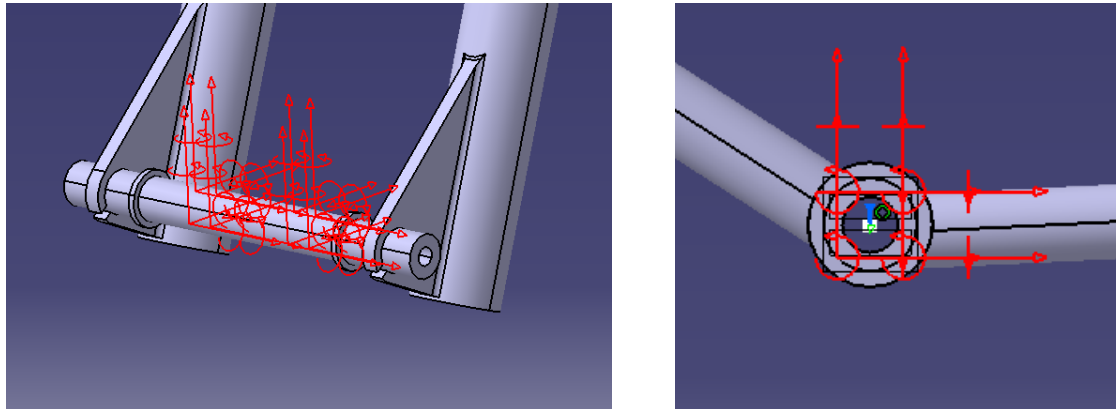


Figure 10: Movement restraints (Left: Front axle, right: Rear axle)

- **Loads:** Loads refer to the forces acting on the body. The force on the seat post is 1500N (150kg) in the direction of negative z-axis. Additionally, the force on the rear seat is double the front i.e., 3000N (300kg) in negative z-axis. In total, 4500N is the force given on the frame to simulate three people weighing 150kg each sitting on the tricycle. 150kg is considered because of the factor of safety to emulate the environment with the highest deformation of the frame and to prevent any failures in the design. (Figure 11)

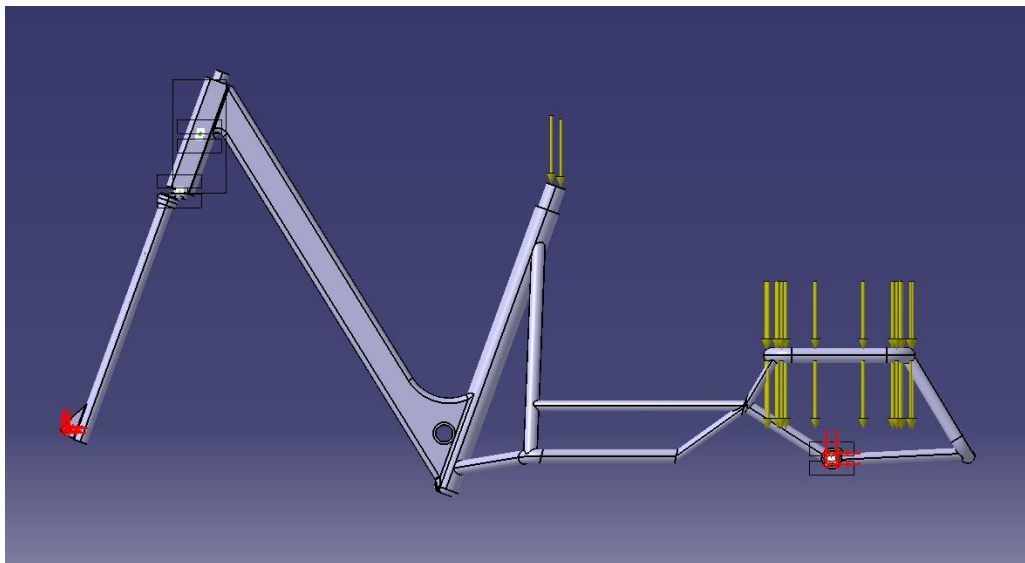


Figure 11: Loads on the frame

With each parameter such as GAC's, mesh size, restraints, loads defined in the workspace, final inspection is a recommended action to check for missing or undefined parameters required for the software to generate accurate results. Figure 12 shows the full assembly with all the parameters defined.

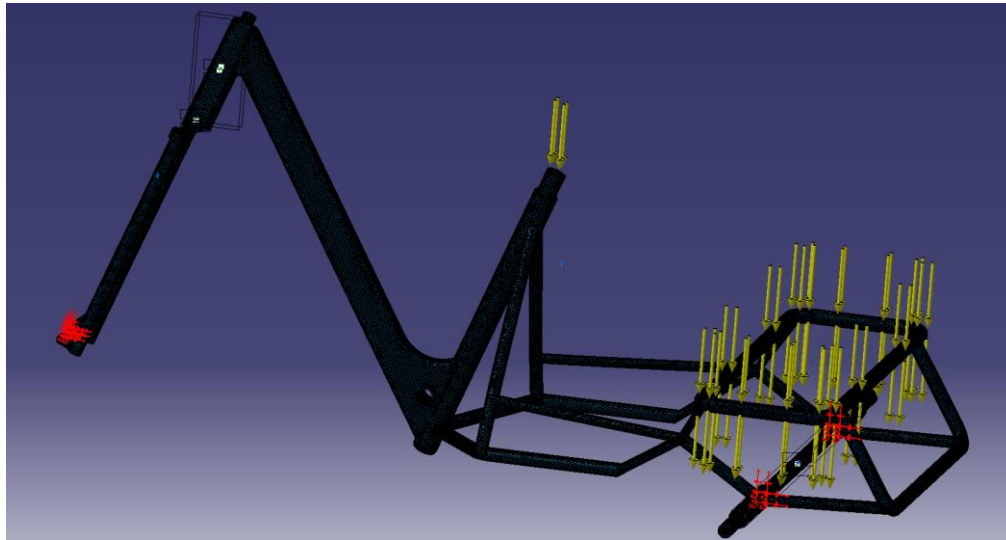


Figure 12: Fully defined analysis model

2.6.4. DESIGN CHANGES POST- INITIAL DESIGN OPTIMISATION RUN

The design optimisation includes refining design of the existing frame. Edge fillets to the corners which has stress concentration points. Addition of shape of material to the frame to help uniform distribution of the forces on the frame.

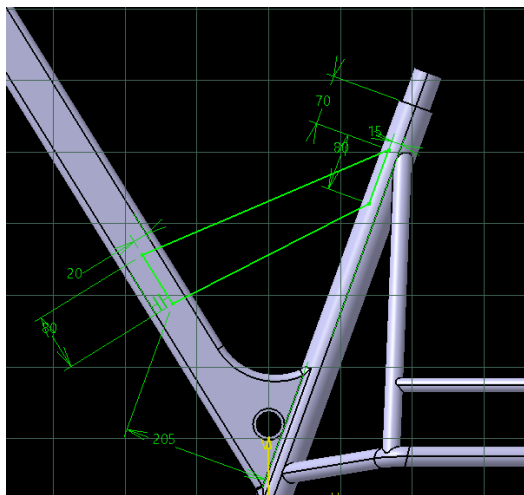


Figure 13: (Left) Added sketch for new material, (Right) Refined new material

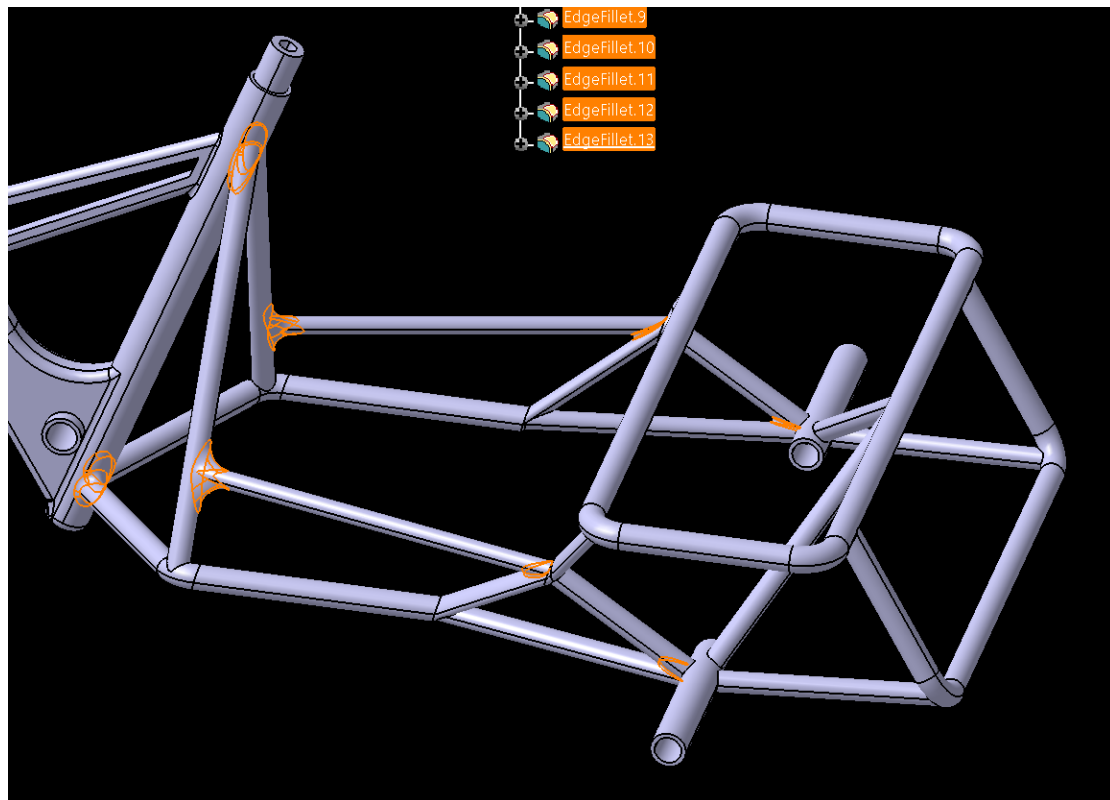


Figure 14: Fillets applied on the frame

The above-mentioned figures 13 and figure 14 shows refinement in the initial design. These are done to rerun the simulation and generate better results. The added material on the frame increases the weight by 2.83kg. Two simulations with two values for the edge fillets are conducted. The first run with the edge fillet of radius 15mm whose results are discussed below.

As the first run did not generate acceptable results, one more simulation was conducted with larger edge fillets of radius 30mm. The results are discussed below

3. RESULTS AND DISCUSSION

3.1. MESH INDEPENDANCY ANALYSIS

Mesh Independancy study					
Mesh Size (mm)	Sag (mm)	Displacement (mm)	Percentage deviation (%)	Von mises Stress (N/m ²)	Time taken
10	8	4.05		4.73E+07	6s
8	6	4.29	5.594405594	5.56E+07	12s
6	5	4.58	6.331877729	6.42E+07	30s
5	4.5	4.72	2.966101695	7.21E+07	1m 6s
5	4	4.72	0	7.21E+07	58s
4	3.6	4.84	2.479338843	8.90E+07	6m 23s
4	3	4.84	0	8.90E+07	5m 3s
3	2.7	4.93	1.825557809	8.01E+07	41m 10s
3	2	4.93	0	8.01E+07	47m 49s

Table 6: Mesh independancy results

The nominal mesh size to start the mesh independancy study was 10mm with a sag value of 8mm, the result of this run is shown in the Table 4 above. As shown, different runs with different mesh size were applied and simulated to find the ideal mesh size.

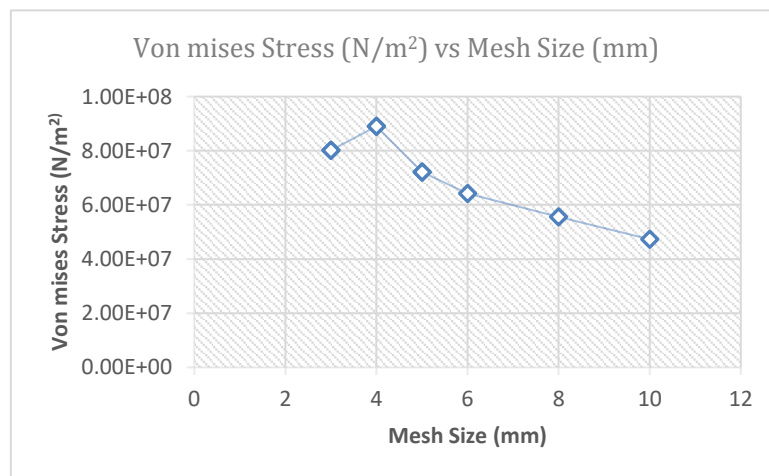


Figure 15: Von-mises stress vs Mesh size graph

Figure 15 shows that mesh size and stress are directly proportional until it reaches a certain point (viz. 3mm), This point shows the ideal mesh size which needs to be used in the analysis as the percentage of deviation between the values generated in the successive run with lower sag size is 1.9% (Refer Table 6).

Simultaneously, Figure 16, follows the same upward trend in the values it represents. The ideal mesh size for the analysis is 3mm and the ideal sag size is 2.7mm. This is chosen because of its lowest percentage of deviation and no change in values when the simulation was run at even lower values in the successive run.

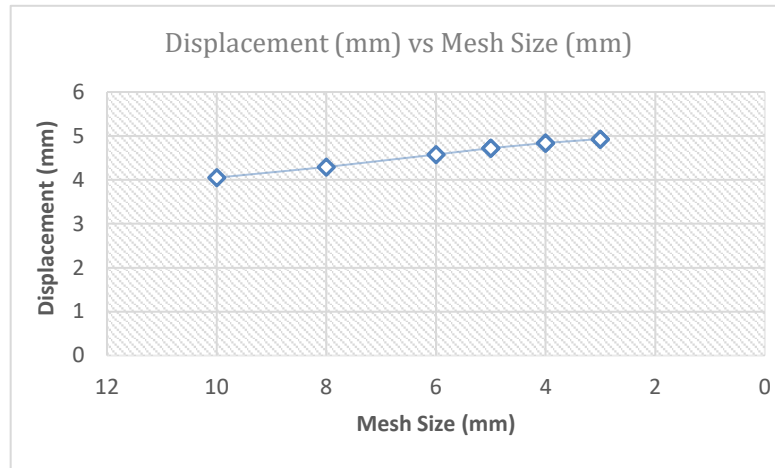


Figure 16: Displacement vs Mesh Size graph

3.2. MATERIAL OPTIMISATION ANALYSIS

Post Mesh independency study, the model is set up for determining the best material that can be used to build the tricycle. Various parameters such as connections between parts, restraints and forces on the frame are defined. This set of parameters is identical to those that are defined during Design optimisation. The ideal material should offer the less deformation under load is lightweight which makes the handling easier. The below table shows the results of Material optimisation study

Material Study					
Material	Displacement (mm)	Von mises Stress (N/m ²)	Principle Stress (N/m ²)	Mass (kg)	Cost of 1m of 25mmØ rod in £ (ex VAT)*
Aluminium 6061 alloy	4.93	7.98E+07	7.94E+07	25.749	15.49
Mild Steel	1.8	7.98E+07	7.88E+07	74.861	20.1
Carbon Fibre Reinforced Polymer	2.71	7.93E+07	8.18E+07	16.689	33.65

*subject to change

Table 7: Material study results with Cost in GBP

Inspecting Table 7, we can see that the deformation of the model when built with mild steel produces very less displacement in the frame. However, the overall weight is significantly high due to its high density (refer Table 5). Cancelling out Mild steel we are left with two exceptionally reliable materials. CFRP shows minimal displacement and keep the assembly just around 17kg. In contrast with Aluminium 6061 which has the largest displacement among the three materials, CFRP is twice the cost of Aluminium 6061. Aluminium 6061 is chosen as the material for the tricycle because it has acceptable displacement value with more uniformity in stress dissipation. It is also lightweight and is the cheapest material to buy among the three.

3.3. DESIGN OPTIMISATION

Design optimisation of the CAD model with pre-set parameters of load and constraints, the objective is to decrease the displacement and alter the design in a way that stress dissipates uniformly onto the frame from the concentration points.

To understand the optimisation technique, the following comparison is necessary.

- CAD model:** The figures 17 and figure 18 below show the initial CAD model (ICM) and the optimised CAD model (OCM). The constraints applied during the analysis forms the basis of the environment. The difference between the two can be seen in the addition of material from the seat post to the frame. This design was chosen so that when the load was applied, the added material would act like as a support to the seat post and reduce displacement of the frame which is discussed in detail below. The material has a hollow design which is intended to save weight without affecting the function for which it is added. Consequently, the fillets applied to the connections from the seat post to the rear frame have been increased in diameter. The purpose of this is to reduce the impact of stress concentration points on the frame under load.

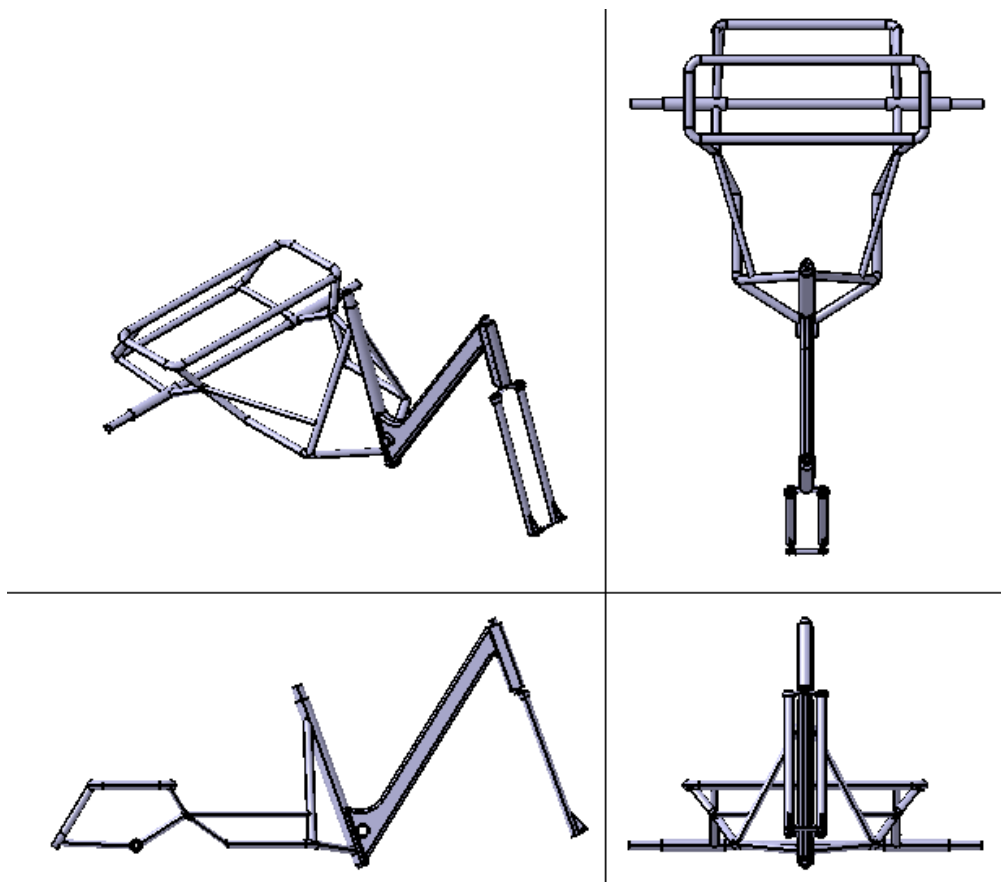


Figure 17: Initial CAD model

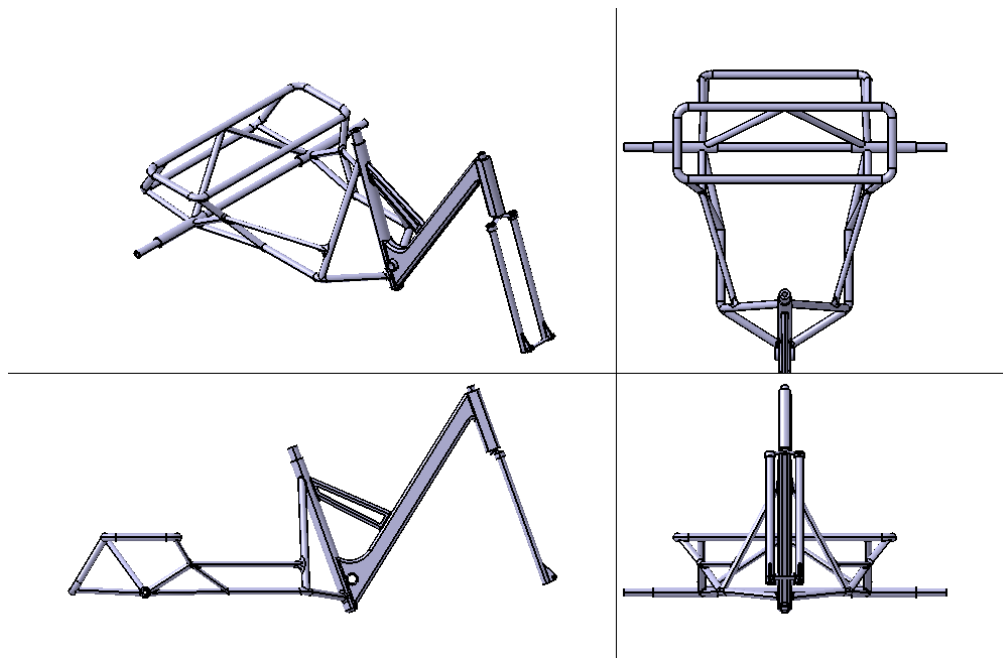


Figure 18: Optimised CAD model

3.3.1. FIRST SIMULATION RESULTS

- **Displacement:** Displacement refers to how much the frame bends when put under load. The presence of the added material helps to reduce deformation as seen in the figures below (19 & 20)

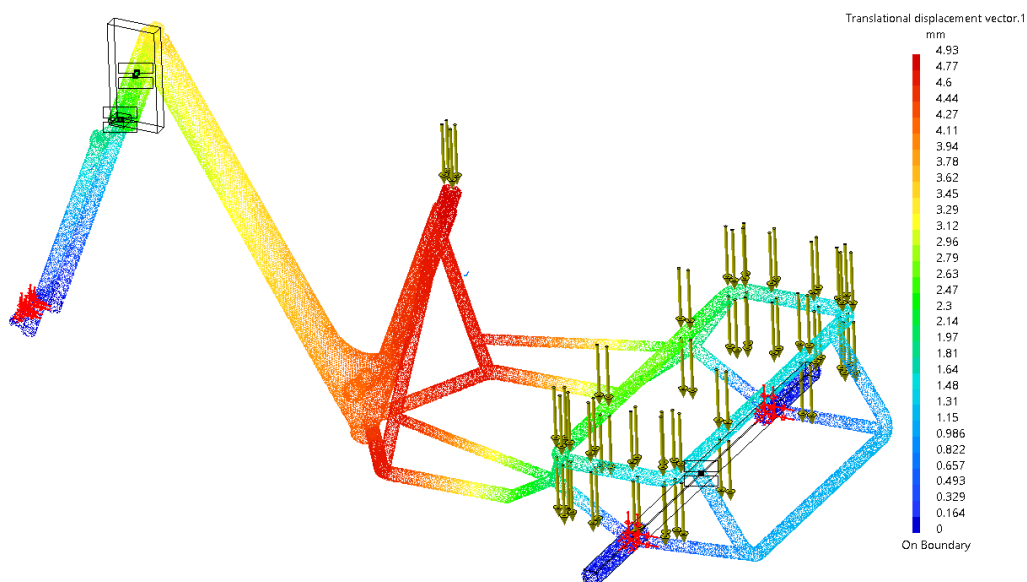


Figure 19: ICM displacement solution

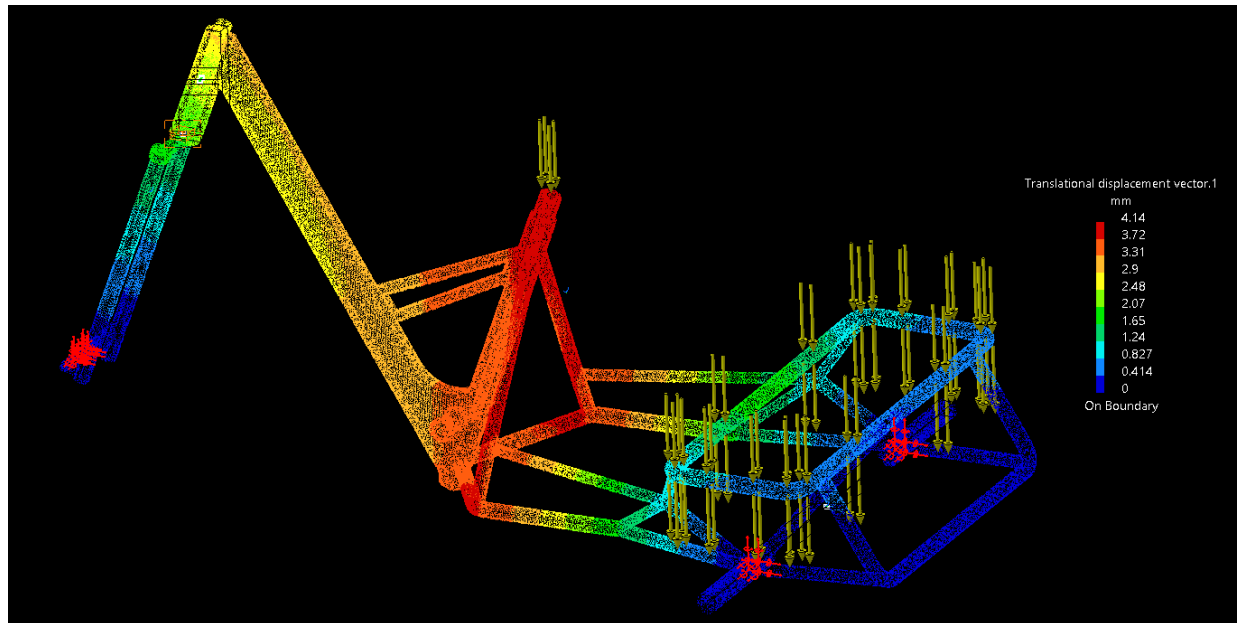


Figure 20: OCM1 displacement solution

The displacement of the frame was reduced from 4.93mm to 4.14mm which translates to a decrease in 16.63%. The Optimised CAD Model 1 is basis for the Optimised CAD Model 2 which was used to further improve the results

- **Von-Mises Stress:** The state of stress at a point in the body is a crucial factor to be analysed. The stress on the frame were at the highest behind the seat post as shown in figure 21.

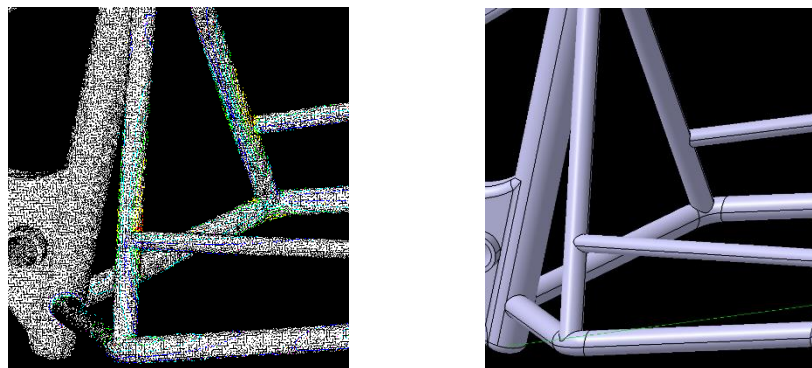


Figure 21: (Left) ICM stress points, (right) ICM model

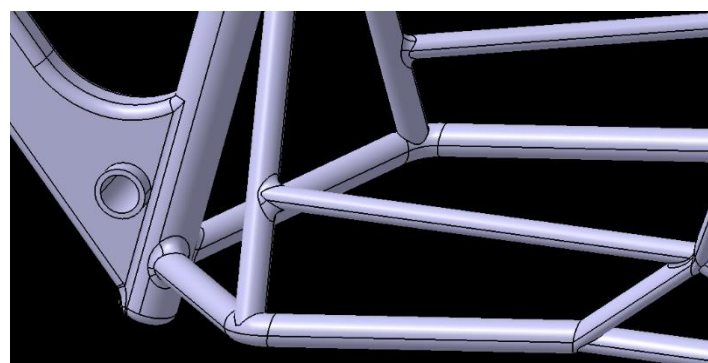


Figure 22: Optimised stress points with fillets

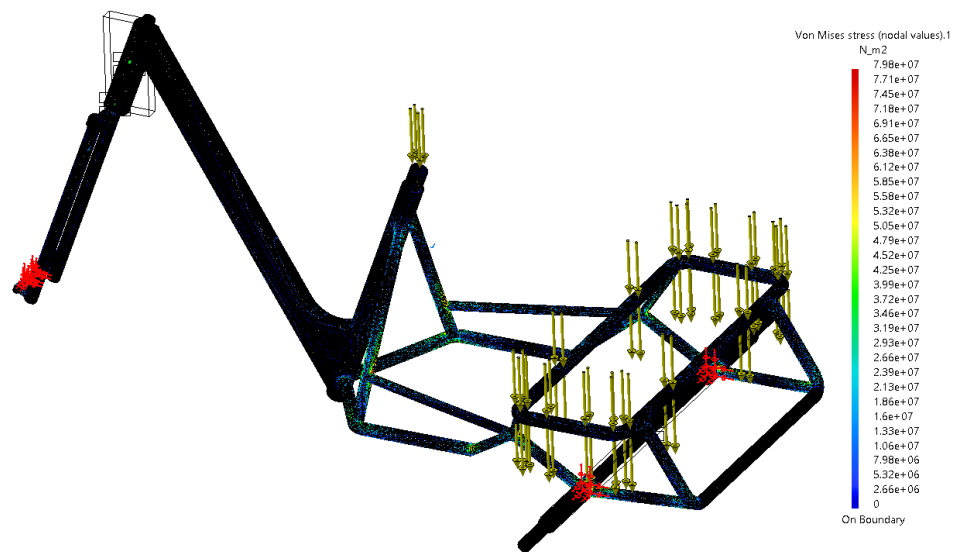


Figure 23: ICM Von-Mises stress solution

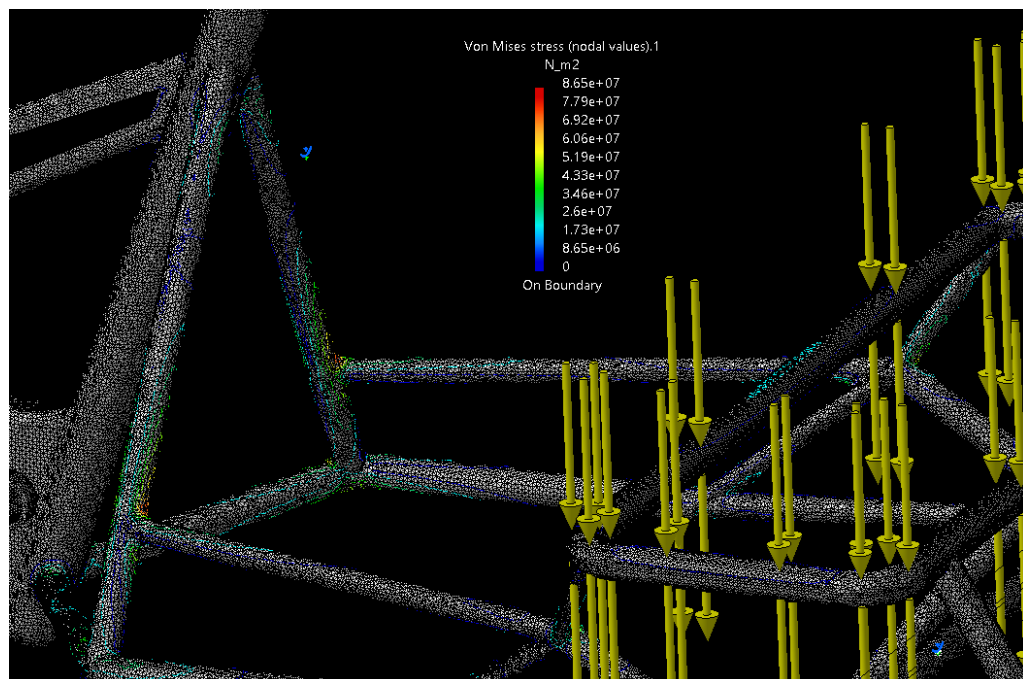


Figure 24: OCM1 Von-Mises Stress solution

Comparing the results of Von-mises stress, the stress has increased by 8.72%. The expected results for this simulation were such that it reduces the value of the stress, however, the next simulation was conducted with bigger edge fillet radius which is discussed in the second simulation results' section below.

- **Principal stress:** In contrast to the Von-mises stress, Principal stress represents the maximum and minimum stress acting on the structure. ICM had the maximum stress at $7.94 \times 10^7 \text{ N/m}^2$ (Figure 25). Refinement of the design leads us to the OCM1 in which the principal stress reduced to $6.05 \times 10^7 \text{ N/m}^2$ which is about 24% (Figure 26).

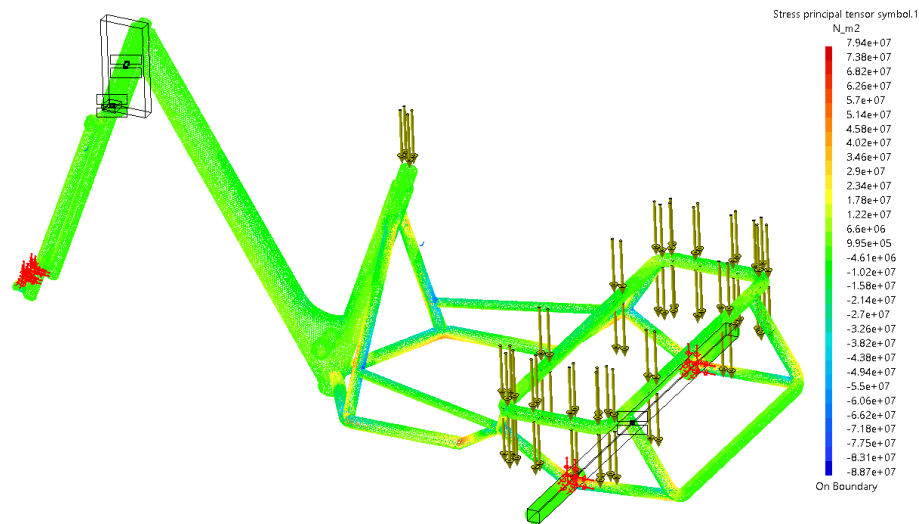


Figure 25: ICM Principal stress solution

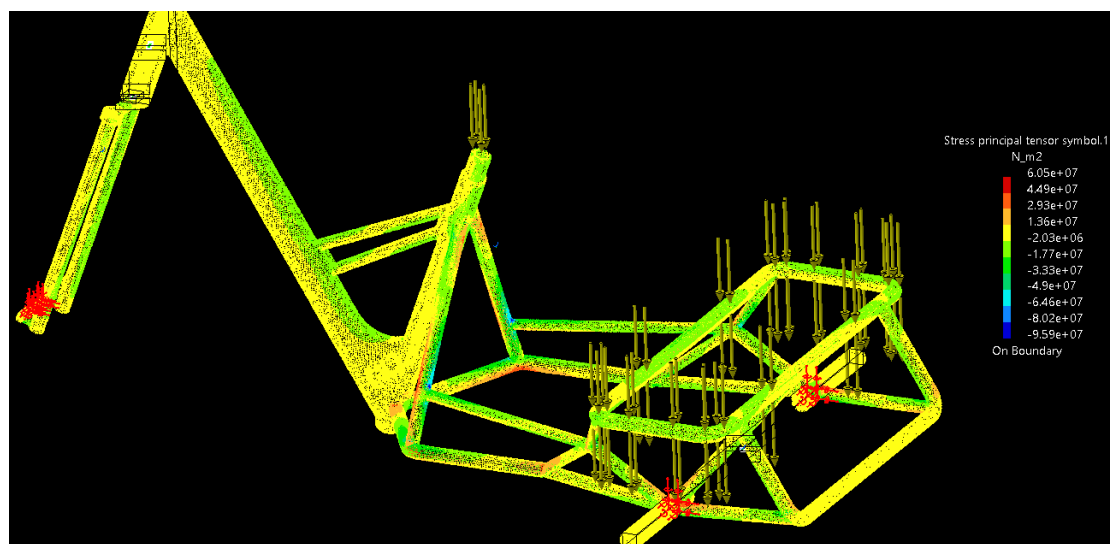


Figure 26: OCM1 Principal stress solution

3.3.2. SECOND SIMULATION RESULTS

- Displacement:

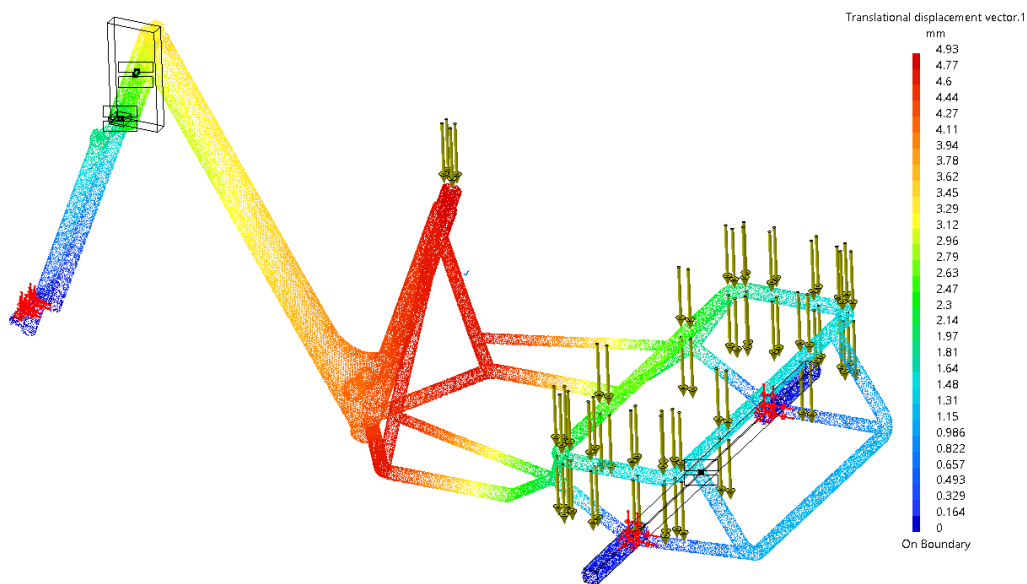


Figure 27: ICM displacement solution

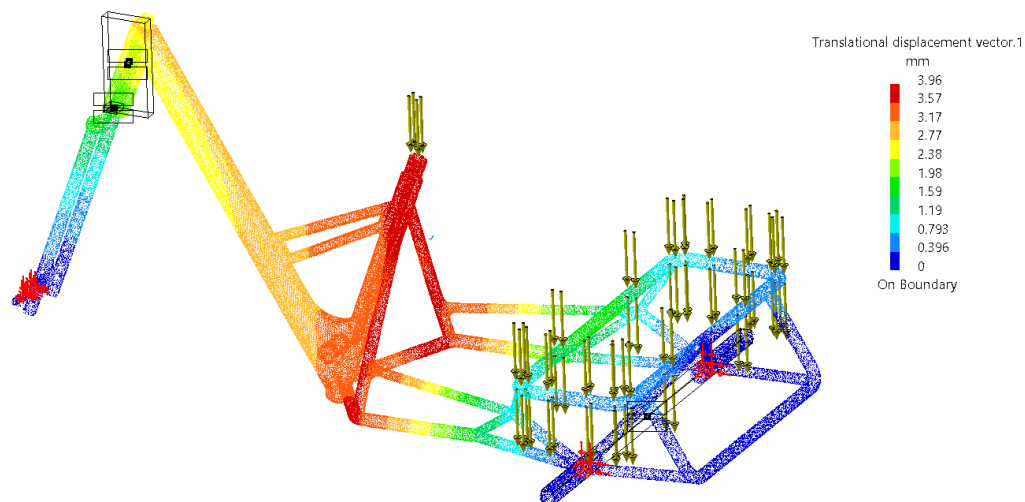


Figure 28: OCM2 displacement solution

The addition of the larger fillet radius reduces the displacement by 20%. Inspecting the optimised solution figure (i.e., Figure 28), the value decreases from 4.93mm to 3.96mm compared to the value of OCM1 which is 4.14mm. Furthermore, it also helps the displacement force to flow to the front of the frame in a more uniformed way.

- **Von-Mises Stress:** Figure 29 and figure 30 represents the difference in the uniformity of the flow of the stress on the mainframe. Von-mises stress is a reliable factor used in optimisation. The Von-mises stress in the OCM2 is decreased by 9% to that of the Initial model.

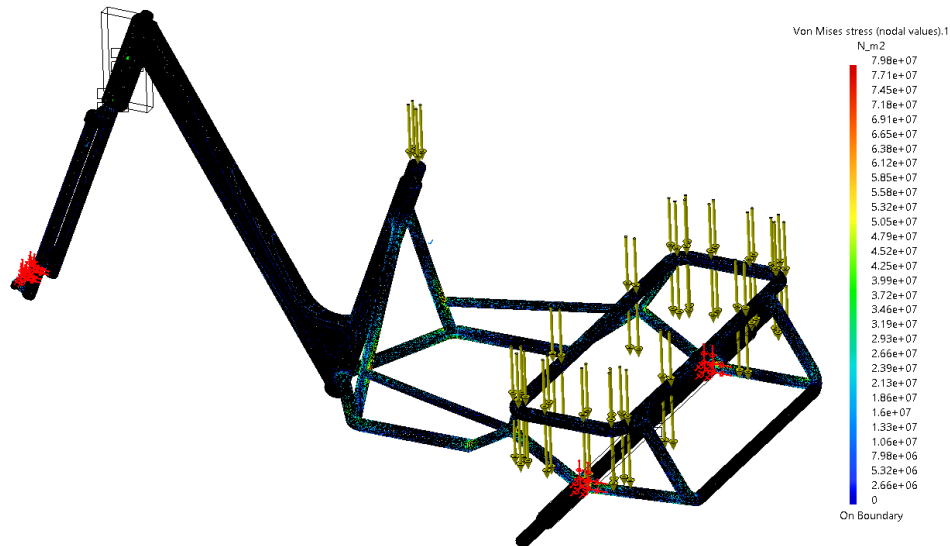


Figure 29: ICM Von-Mises stress solution

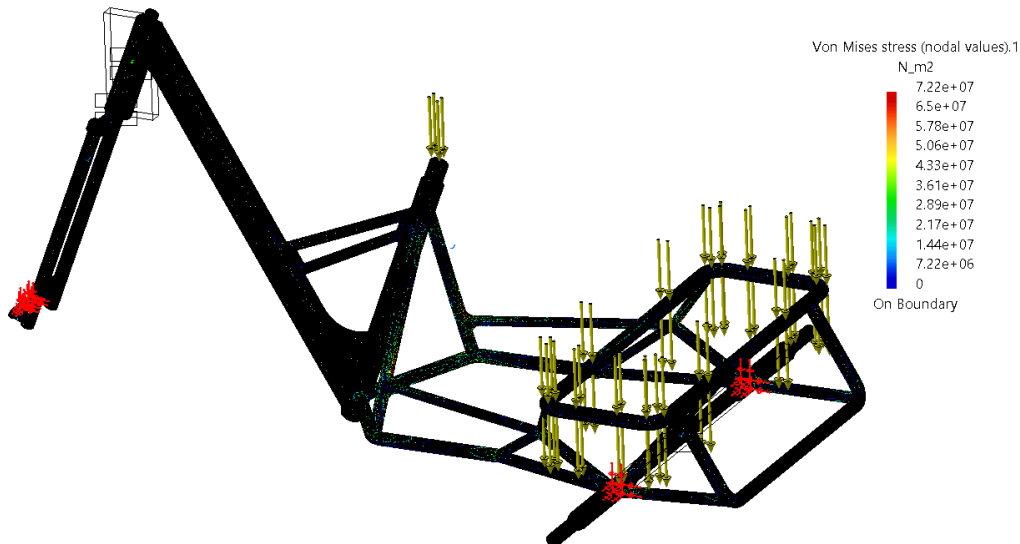


Figure 30: OCM2 Von-Mises Stress solution

- Principal stress:** In contrast to the Von-mises stress, Principal stress represents the maximum and minimum stress acting on the structure. ICM had the maximum stress at $7.94 \times 10^7 \text{ N/m}^2$. Refinement of the design leads us to the OCM2 model within which the stress has decreased by 30% (refer figure 31 & figure 32), which proves that the refinement is effective.

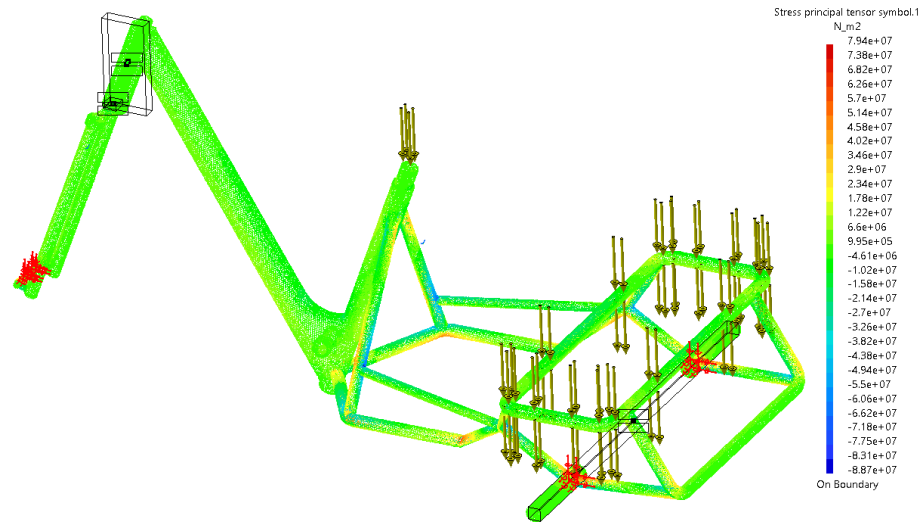


Figure 31: ICM Principal stress solution

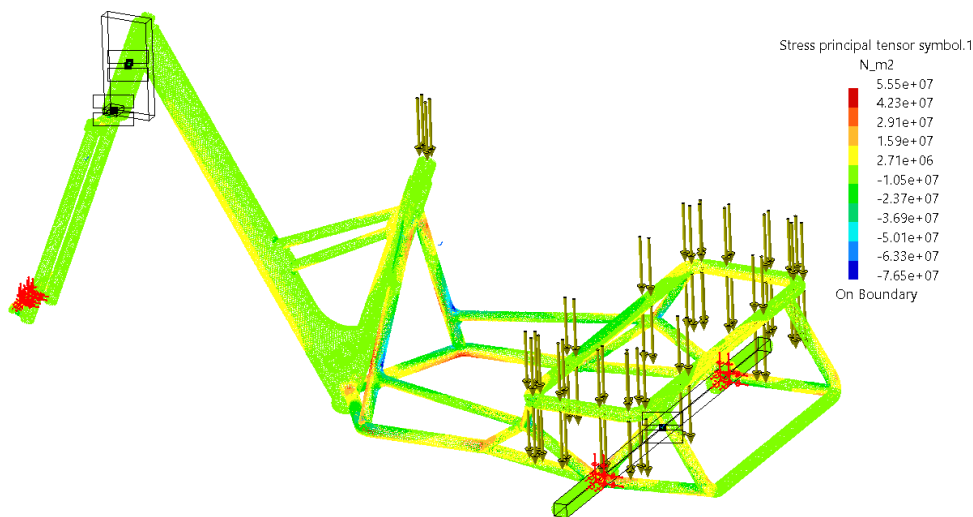


Figure 32: OCM2 Principal stress solution

3.4. FINAL COMPARISON OF THE OPTIMISATION

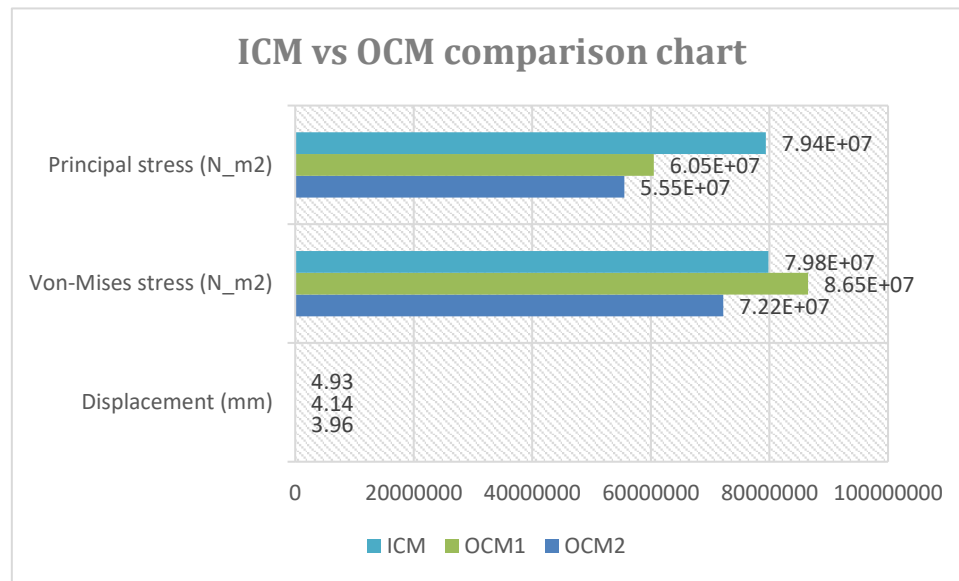


Figure 33: ICM and OCM comparison graph

Figure 33 shows the improvement in displacement, Principal and Von-Mises stress from the Initial CAD Model to the Optimised CAD Model. Successfully meeting all the requirements of the product design specification, the final assembly (figure 34) is with the modified design is a potential basis to develop an electric tricycle.

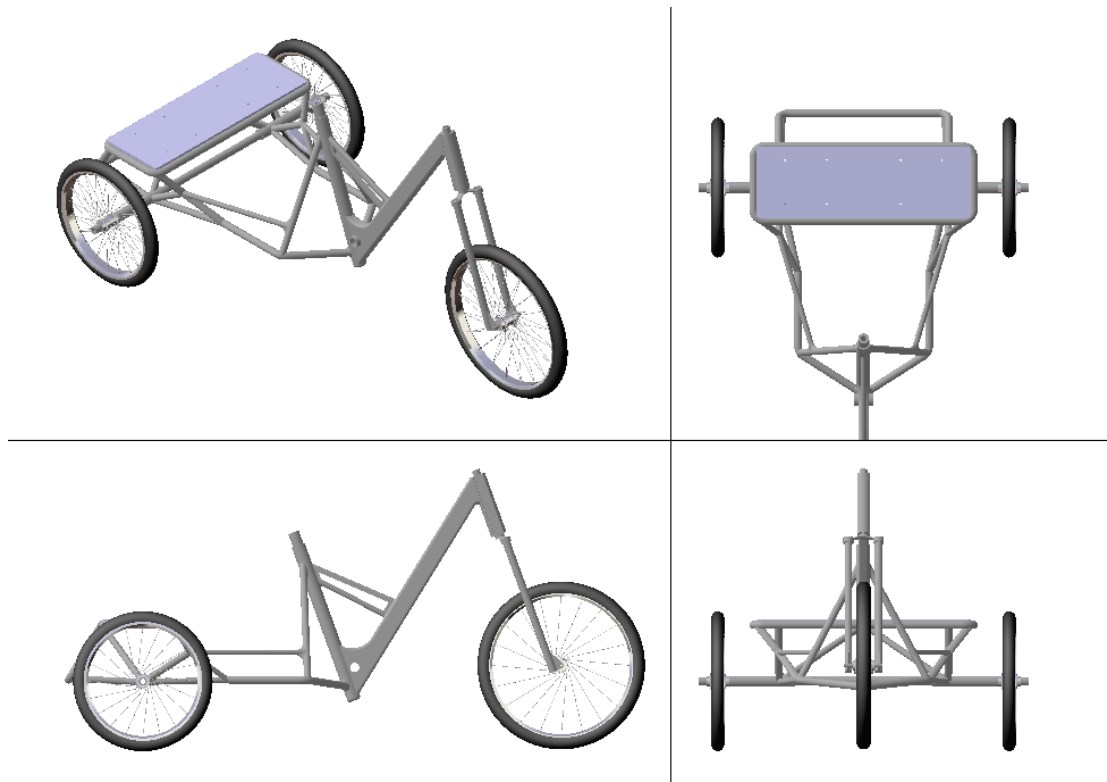


Figure 34: Fully assembled tricycle

4. CONCLUSION

A design study and development of an electric tricycle was accomplished. Undertaking the project beginning with the literature review which formed the basis of the product design specification of the electric tricycle and further designing a 3D model in CATIA V5. Following the design process i.e., mesh independency study, with which we identified that using 3mm as mesh size and 2.7mm sag size in the static analysis generated accurate results well within the accepted time period. Material optimisation of the frame resulted in using Aluminium 6061 alloy material as it acceptable displacement while keeping the structure lightweight and affordable. Design optimisation using determined parameters was conducted. Constraints on the frame and applying loads on it with an overall value of 450kg. The initial values of displacement, von-mises stress and principal stress was noted and the design was modified through which we got the Optimised Tricycle Design. Inspecting the design optimisation results, we come to know that the OCM2 (Optimised CAD Model 2) is better than ICM (Initial CAD Model). OCM2 had 20% less displacement in the frame structure at 3.96mm. Additionally, the von-mises and principal stress on the frame structure was decreased by 9% and 30% respectively. The value of these analysis is necessary to compare the efficiency of the design with the market standards. Through the analysis, we gained insight into the design and development process of the product. It was determined that optimizing the design was crucial for improving the overall design while staying within the constraints.

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APPENDICES

A. GANTT CHART

Design of an electric 3-wheeler (adult) tricycle for countryside



Figure 35: Gantt chart with periods of planned and actual work done

B. FIGURES

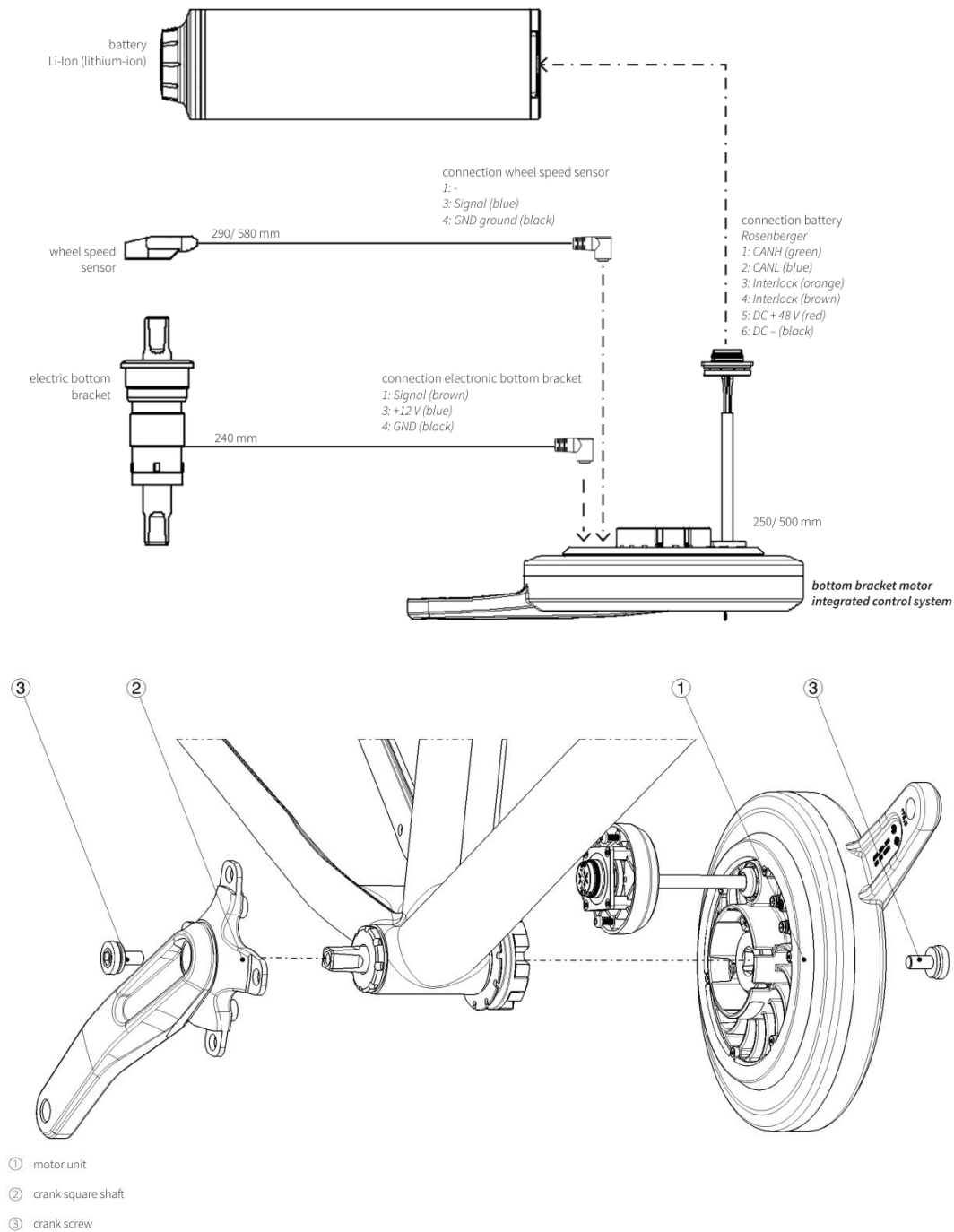


Figure 36: Pendix eDrive installation drawings (Top: Wiring diagram, bottom: drive system assembly)

C. BILL OF MATERIALS

	Part Number	Title	Quantity	Remarks	Source Link
1	541385	Front tyres: Schwalbe Black Jack Bike Tyre 24x2.1	1	24 x 2.1 Inch [609.6 x 53.34 mm]	https://www.halfords.co.uk/product/schwalbe-black-jack-24x2-1-tyre-541385
2	541377	Rear tyres: Schwalbe Black Jack Bike Tyre 20x1.90	2	20 x 1.90 Inch [508 x 48.26 mm]	https://www.halfords.co.uk/product/schwalbe-black-jack-20x1-90-tyre-541377
3	161616	Tubes: Schwalbe Presta Bike Inner Tube - 16" - 26"	3	compatible with the tyre size	https://www.halfords.co.uk/product/schwalbe-presta-bike-inner-tube-16-26-161616
4	479598	Handlebar: XLC HB-C08 Comp Ergo-Bar, 600x31.8mm	1	Diameter: 31.8mm and Length: 600mm	https://www.halfords.co.uk/product/xlc-hb-c08-comp-ergo-bar-479598
5	357335	Stem: Halfords Adjustable Alloy Stem, 25.4mm	1	(d x l) 25.4mm x 110mm, Aluminium 6061 Alloy	https://www.halfords.co.uk/product/halfords-adjustable-alloy-stem-357335
6	382747	Bell: Adie Bike Bell	1		Adie Bike Bell
7	356105	Mirror: Halfords 3" Bike Mirror	1		Halfords 3" Bike
8	351343	Chain: Shimano CN-HG40 6, 7 and 8 Speed 116 Link Bike Chain with Connecting Link	1	116 links, Steel	Shimano CN-HG
9	778308	Gear Shifter: SRAM MRX Comp 7 Speed Rear Gear Shifter	1		SRAM MRX Com
10	279304	Derailleur: RD-TY300 6/7-speed direct-mount rear derailleur	1	Steel, Long Cage size	RD-TY300 6/7-sp
11	248364	Gear Cassette: Halfords 7 Speed Freewheel 14-28T	1	7 speed, 14-28T size	Halfords 7 Spee
12	SO01201075_BLAC1-UK-XHF	Front seat: Universal Backrest Saddle, Wide Comfort Cruiser Tricycle Bike Bicycle Saddle Seat Pad With Back Rest	1	30 x 24 x 33cm (L x W x H), Leather	Universal Backre
13	TG124-foam-M	Rear seats: Drift Trike Go Folding Seat Cushion with PU Leather Density Foam Comfortable for Kart ATV Buggy Car Saddle Replacement S Size	2	3.75kg, 36 x 41 x 30cm (L x W x H)	Drift Trike Go Fo
14	8262T	M8 screws: EASYFIX A2 STAINLESS STEEL SET SCREWS M8 X 30MM 10 PACK	4	Stainless steel, 30mm length	Easyfix A2 Stain
15	eDrive500	Motor & battery system: Pendix eDrive500	1	refer Product Design Specification	Find a dealer (p